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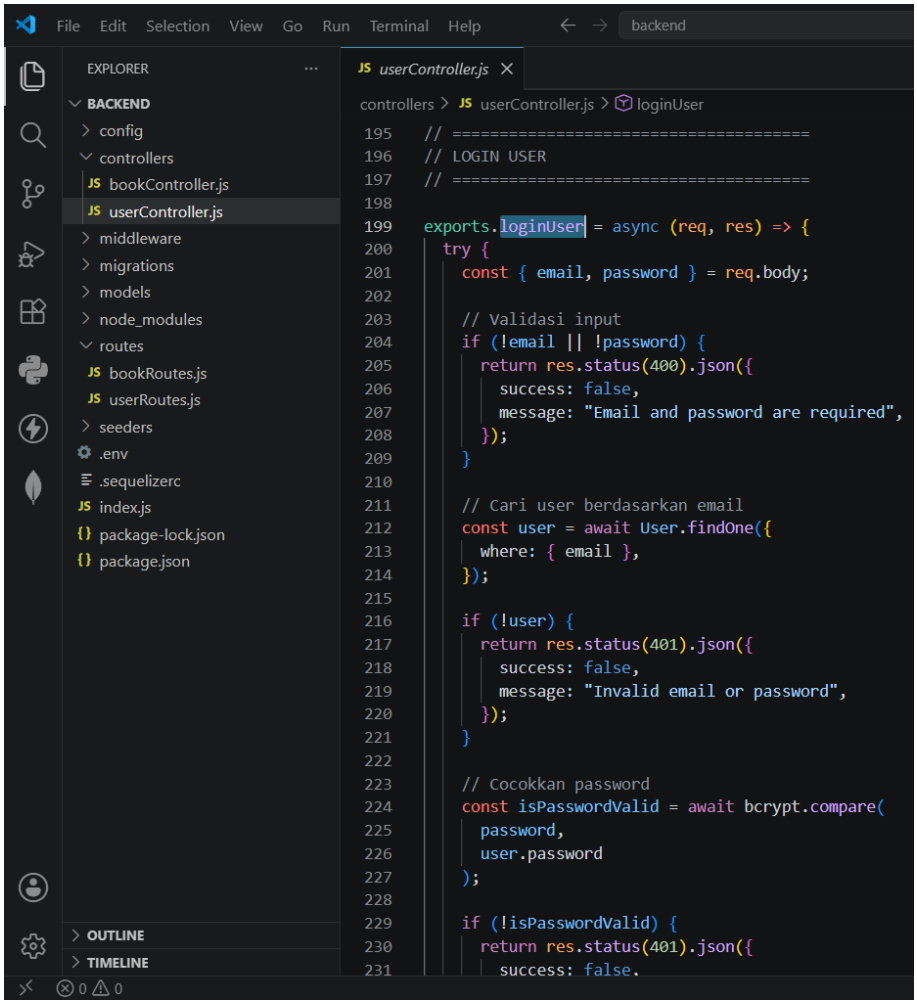
Kelas : TI-24-KA

Mata Kuliah : Lab. Pemrograman Web

BAB VIII AUTHENTICATION AND INTEGRATING

1. Backend Authentication

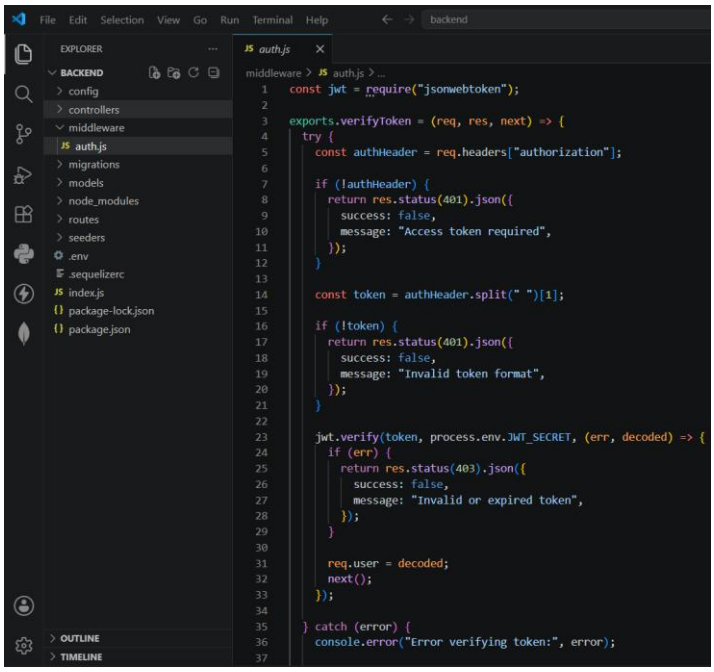
SS 1 backend/controllers/userController.js



Penjelasan:

Menambahkan fungsi login menggunakan bcrypt untuk verifikasi password dan JWT untuk menghasilkan access token yang digunakan pada proses autentikasi.

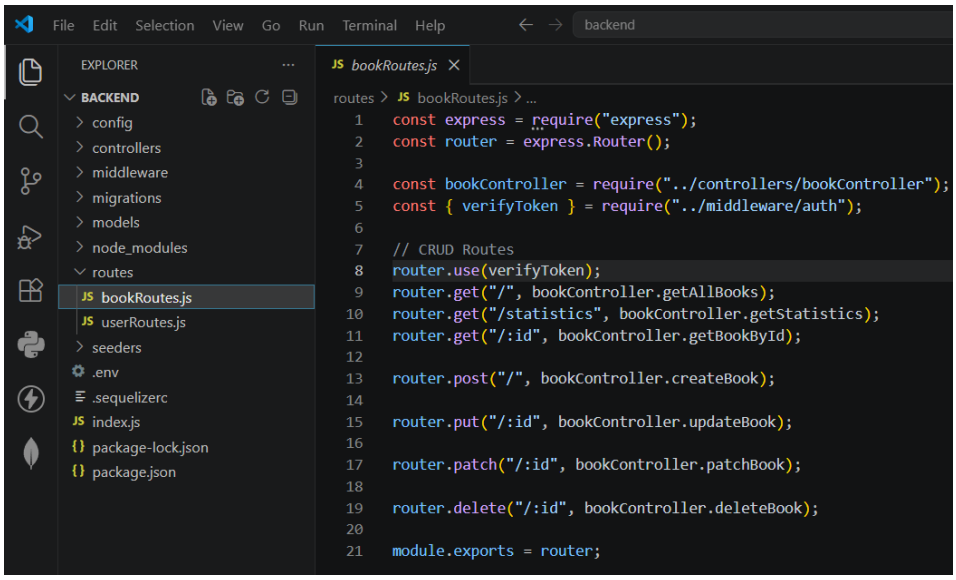
SS 2 backend/middleware/auth.js



Penjelasan:

Middleware verifyToken digunakan untuk memverifikasi JWT sebelum pengguna dapat mengakses endpoint yang bersifat protected.

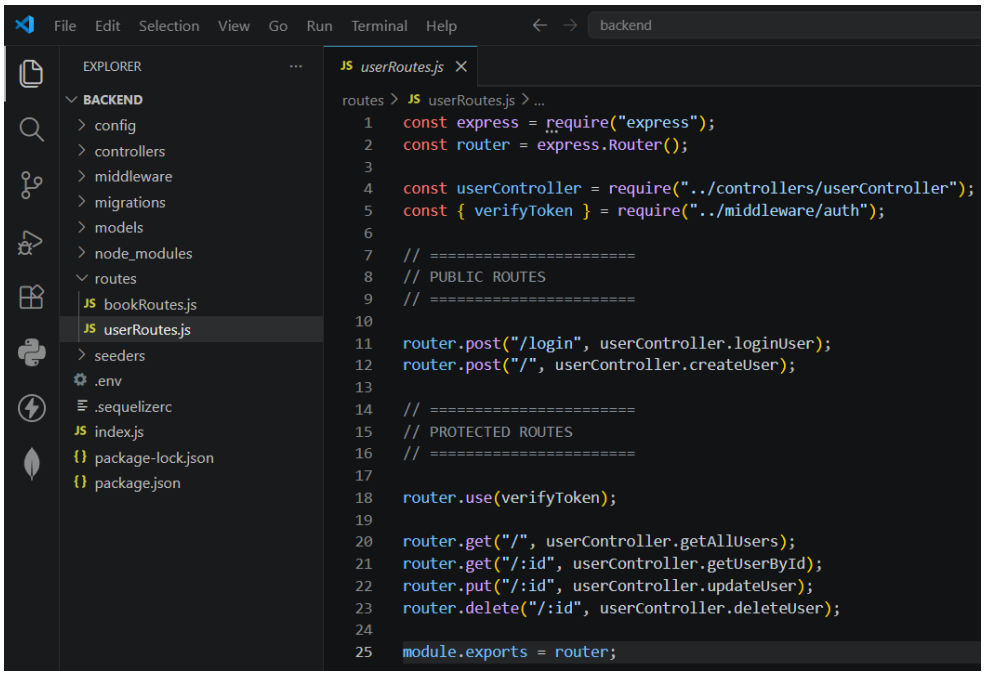
SS 3 backend/routes/bookRoutes.js



Penjelasan:

Seluruh endpoint books diamankan menggunakan middleware verifyToken sehingga hanya pengguna yang telah login yang dapat mengaksesnya.

SS 4 backend/routes/userRoutes.js

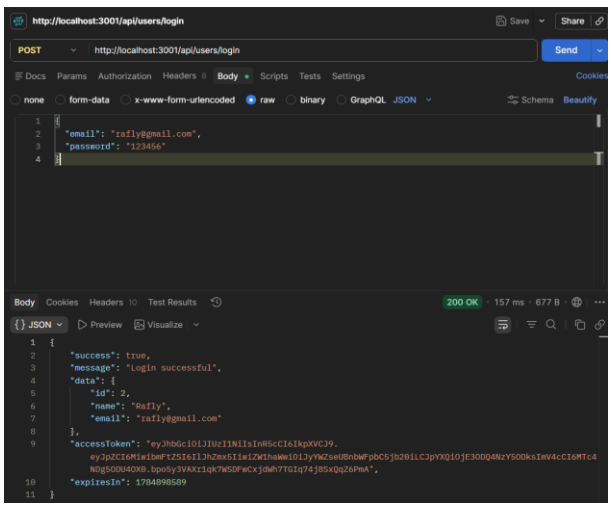


Penjelasan:

Endpoint login dan register bersifat public, sedangkan endpoint CRUD user memerlukan autentikasi JWT.

2. Testing Postman

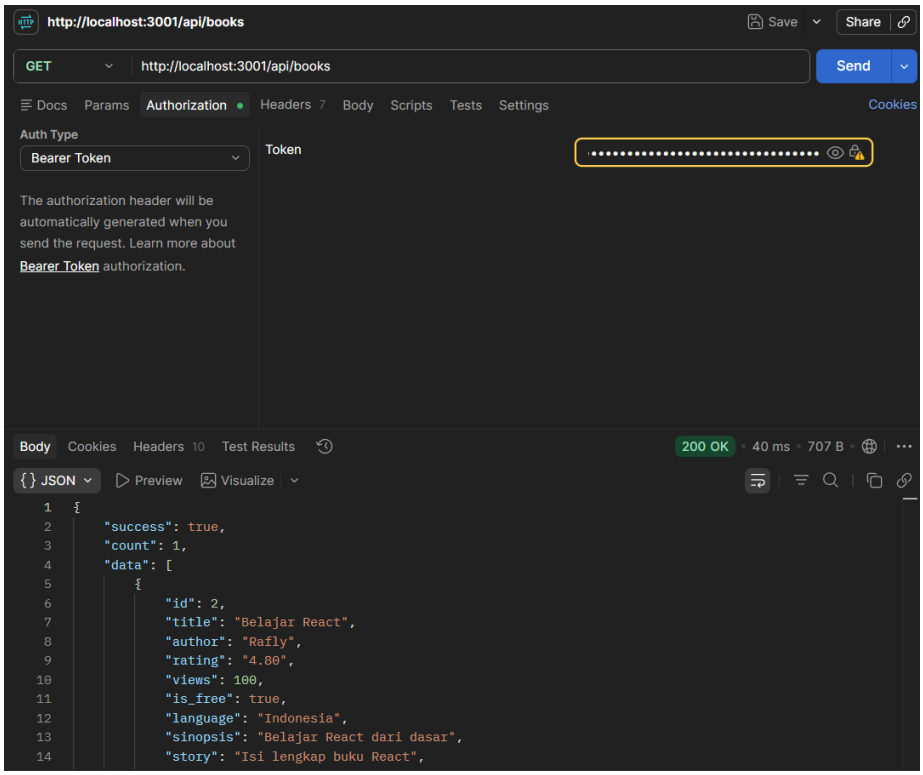
SS 5 POST /api/users/login



Penjelasan:

Pengguna berhasil login dan memperoleh accessToken yang akan digunakan sebagai Bearer Token pada request selanjutnya.

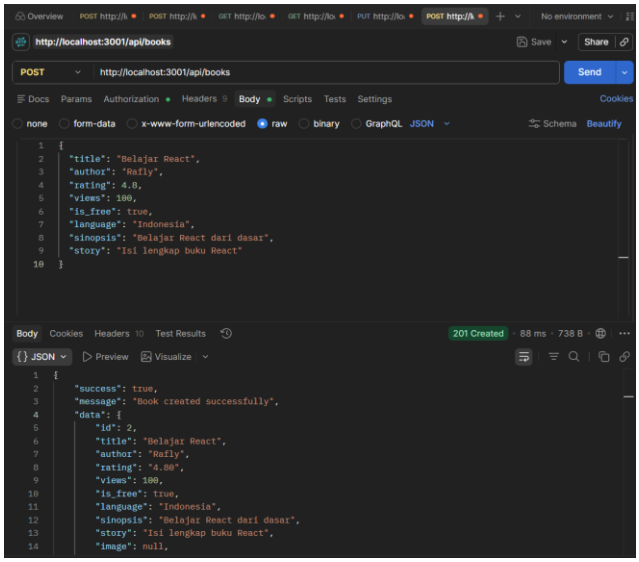
SS 6 GET Books



Penjelasan:

Endpoint books berhasil diakses menggunakan Bearer Token.

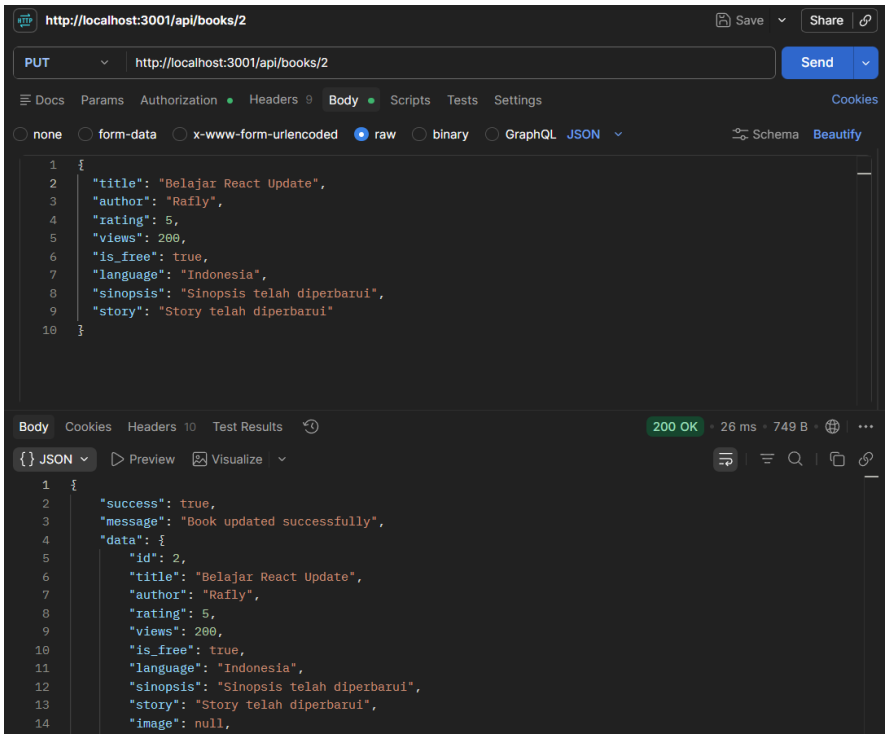
SS 7 POST Book



Penjelasan:

Data buku berhasil ditambahkan ke database melalui REST API.

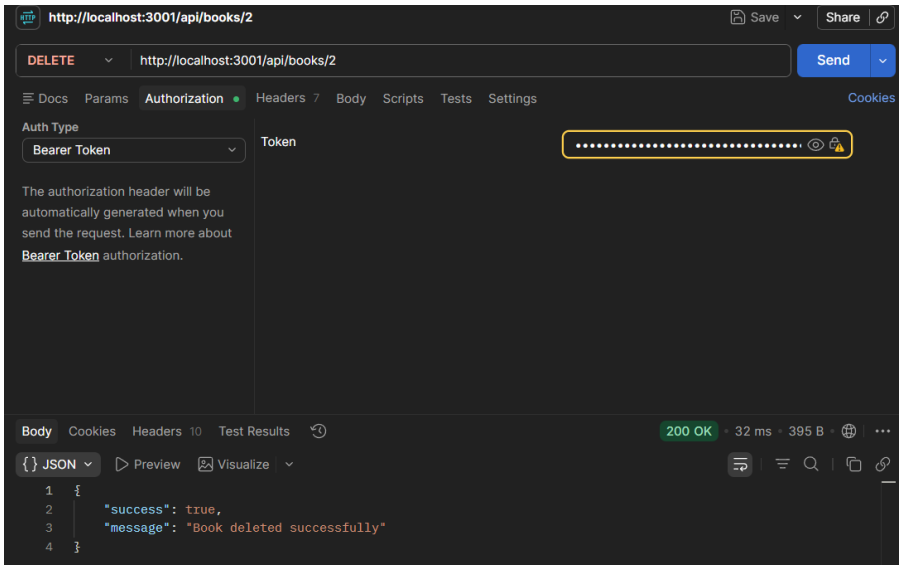
SS 8 PUT Book



Penjelasan:

Data buku berhasil diperbarui menggunakan endpoint update.

SS 9 DELETE Book

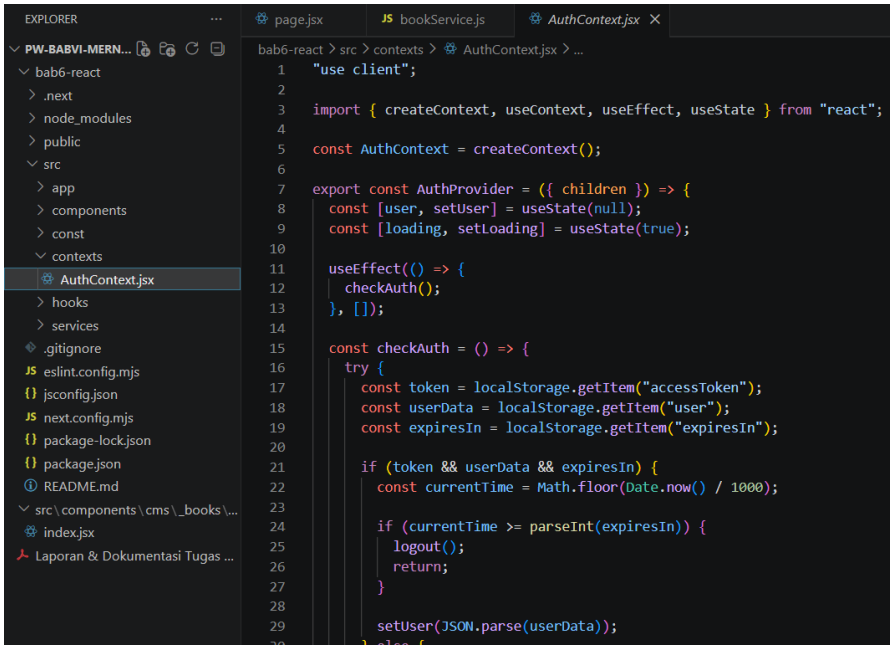


Penjelasan:

Data buku berhasil dihapus melalui endpoint delete.

3. Frontend

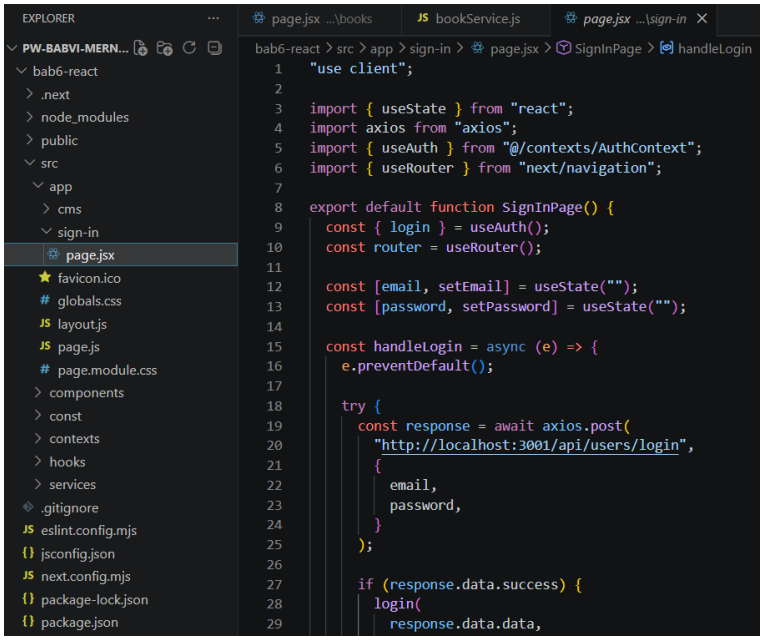
SS 10 src/contexts/AuthContext.jsx



Penjelasan:

AuthContext digunakan untuk mengelola status autentikasi pengguna, menyimpan data login, token, serta mengatur proses login dan logout.

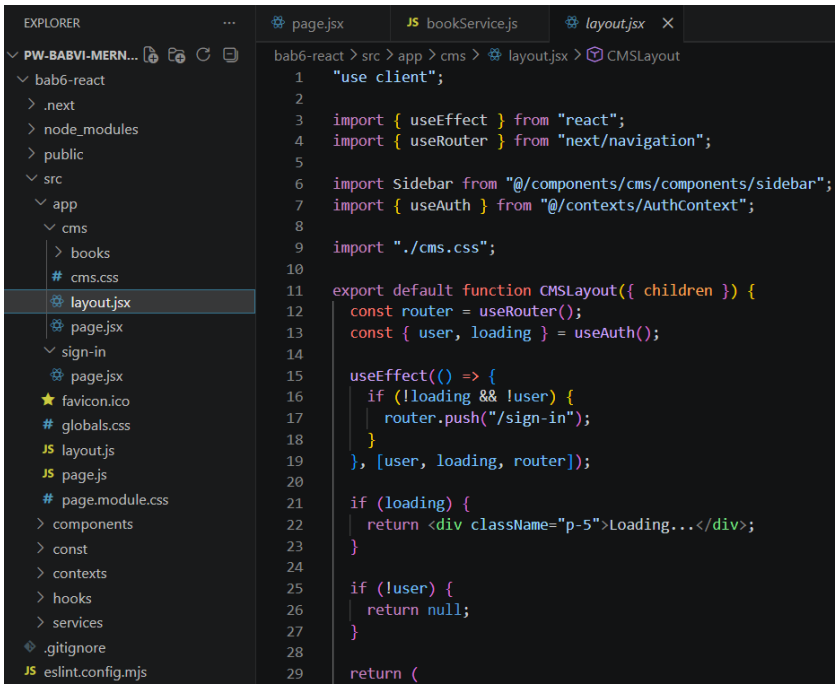
SS 11 src/app/sign-in/page.jsx



Penjelasan:

Halaman Sign In melakukan autentikasi ke backend menggunakan endpoint /api/users/login.

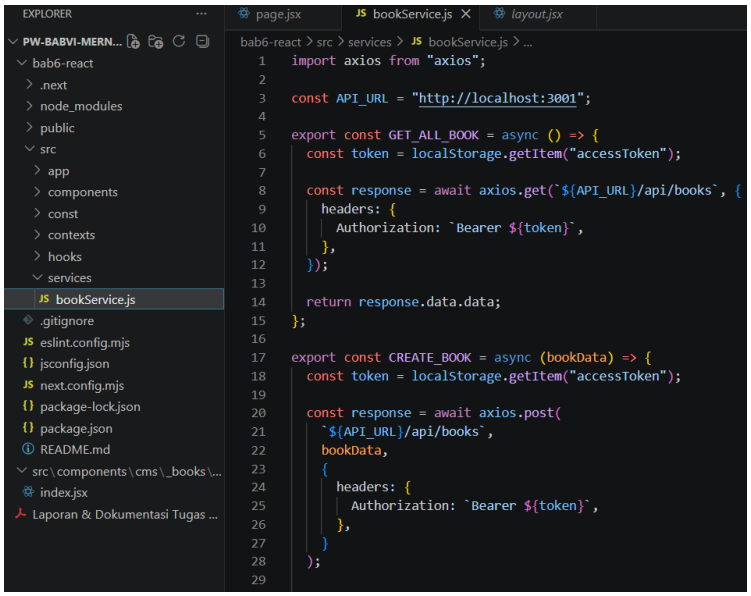
SS 12 src/app/cms/layout.jsx



Penjelasan:

Layout CMS menerapkan protected route sehingga halaman CMS hanya dapat diakses oleh pengguna yang telah login.

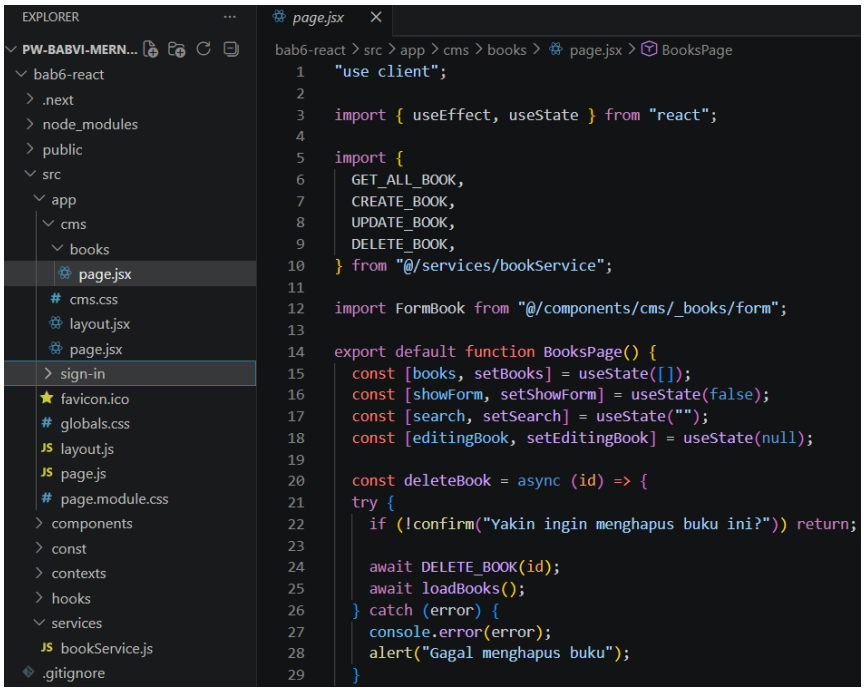
SS 13 src/services/bookService.js



Penjelasan:

File ini berisi fungsi untuk berkomunikasi dengan REST API Books menggunakan Axios.

SS 14 src/app/cms/books/page.jsx

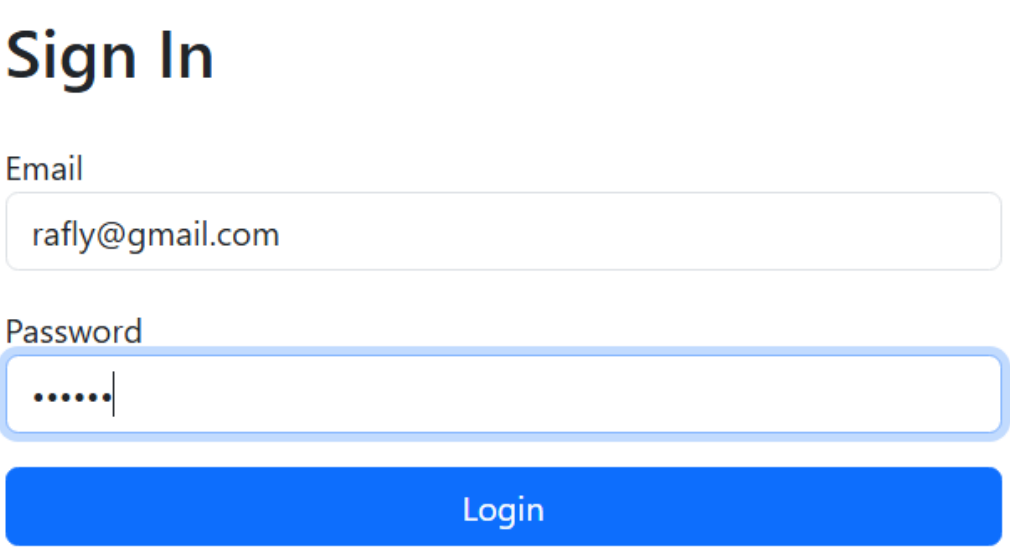


Penjelasan:

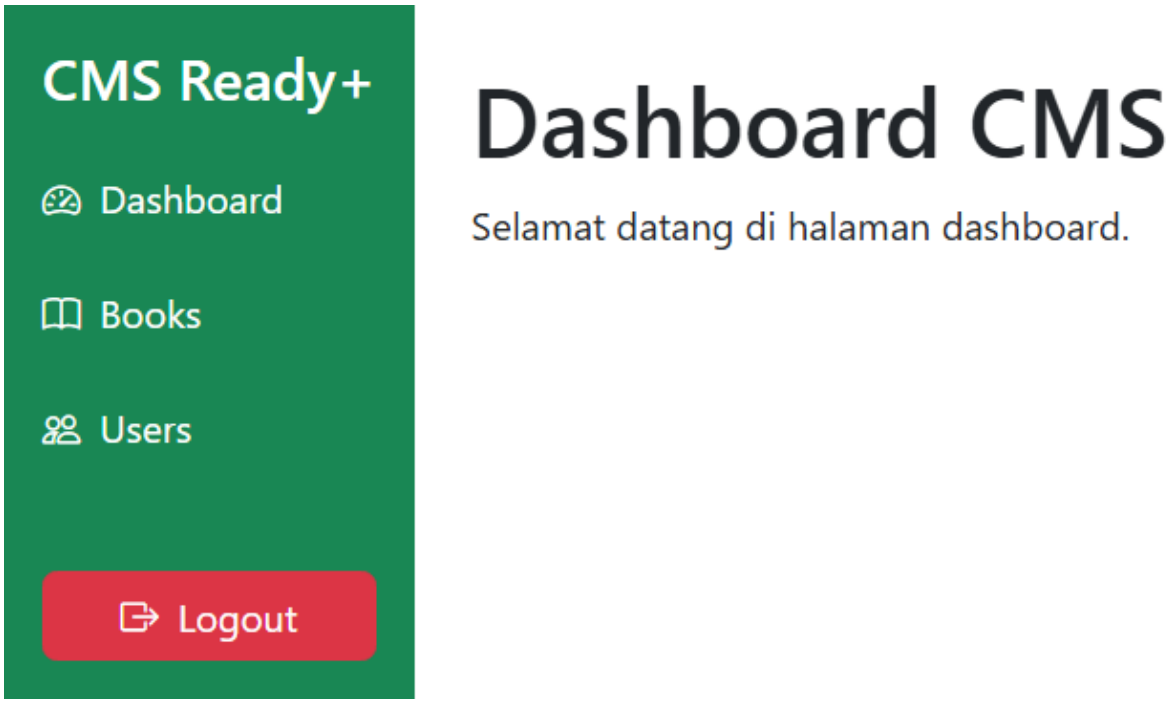
Halaman Books telah diintegrasikan dengan backend sehingga proses menampilkan, menambah, mengubah, dan menghapus data dilakukan melalui REST API.

4. Tampilan Aplikasi

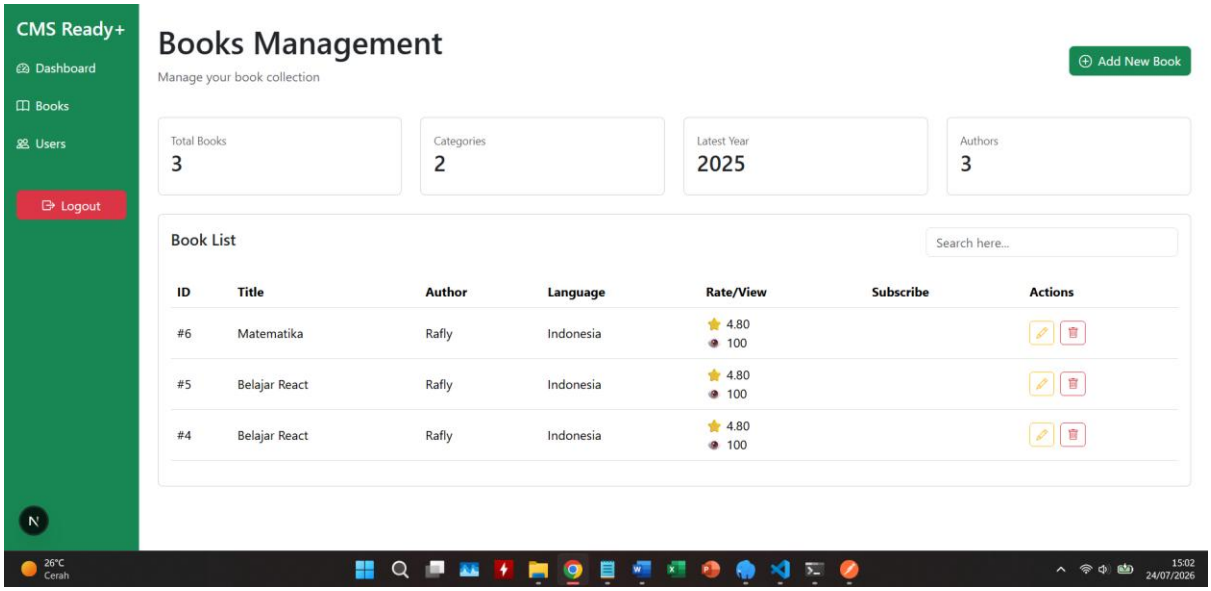
SS 15 Halaman Login



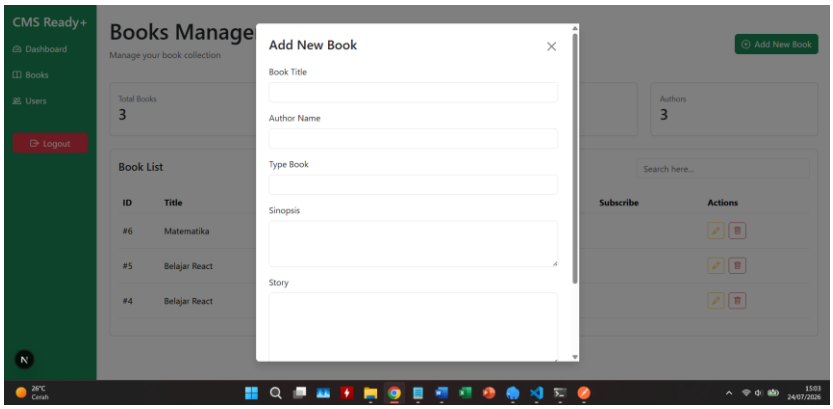
SS 16 Dashboard CMS



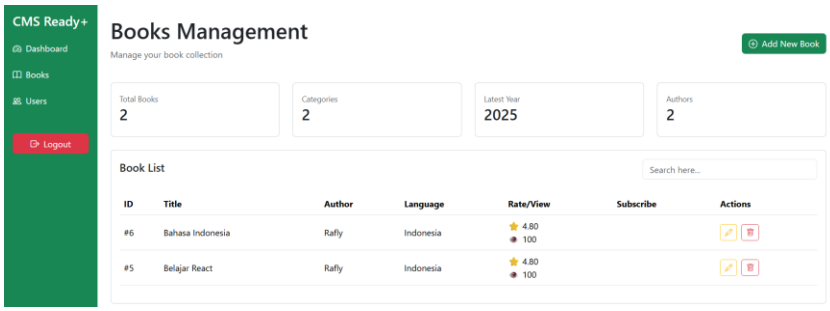
SS 17 Books sebelum Add



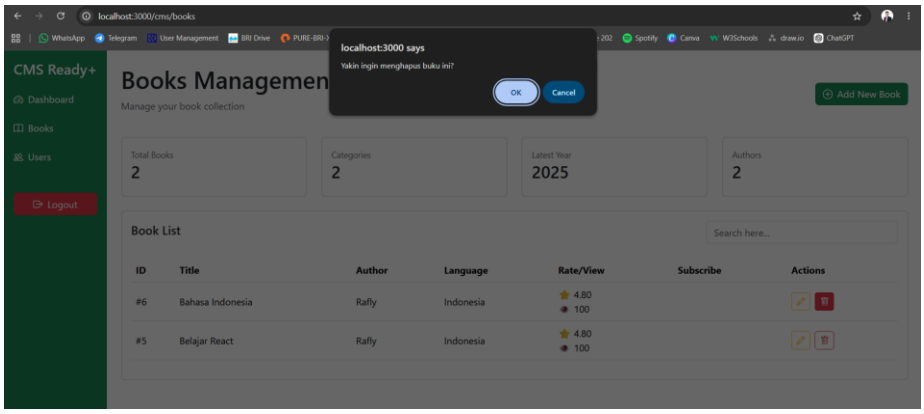
SS 18 Modal Add Book

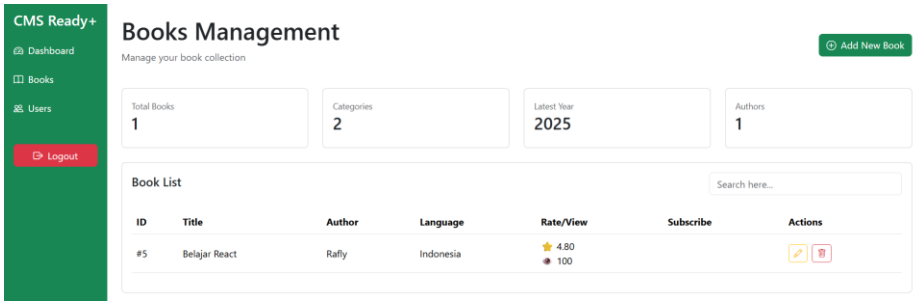


SS 19 Data berhasil diedit



SS 20 Data berhasil dihapus





6. Kesimpulan

Berdasarkan hasil implementasi, proses autentikasi menggunakan JWT berhasil diterapkan pada aplikasi. Frontend Next.js berhasil terintegrasi dengan backend Express.js melalui REST API sehingga proses CRUD Books dapat dilakukan secara dinamis menggunakan database MySQL. Seluruh endpoint protected hanya dapat diakses oleh pengguna yang telah login menggunakan Bearer Token. Proses login, logout, penambahan, perubahan, dan penghapusan data berhasil diuji dan berjalan sesuai dengan yang diharapkan.